

RESPONSES UNDER VOLUNTARY CONTROL

Nervous responses under voluntary control are the opposite of reactions controlled by the ANS. Stimulated by incoming sensory nerve messages, or by conscious thought and intention, the brain's cerebral cortex formulates a central motor plan for a particular movement, and sends out instructions

as motor nerve signals to voluntary muscles. As the movement progresses, it is monitored by sensory endings in the muscles, tendons, and joints. The sensory endings update the cerebellum, so that the cerebral cortex can send corrective nerve signals back to the muscles in order to keep the movement coordinated and on course.

